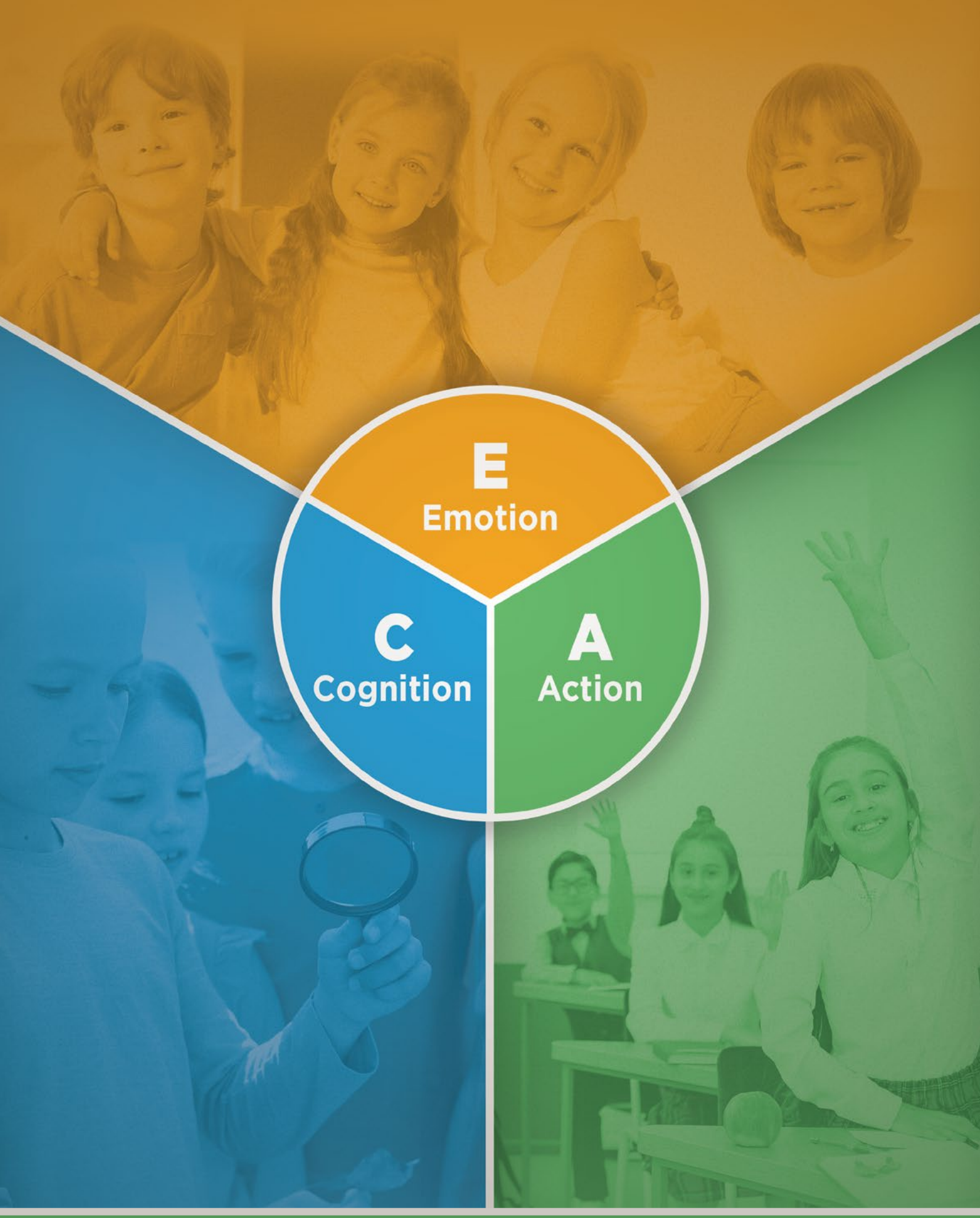


ECA Teacher's Handbook

Author:
Katalin Csizmazia

The full name of the Meseliget Foundation's project is:
Developing a Skill Building Balanced Emotion, Cognition and Action

Publisher: Meseliget Foundation

Project's number: 2025-1-HU01-KA210-SCH-000360331

The project was carried out in collaboration with
the Polish partner organization Gedania1922

Call for proposals (start) date: 2025

AI Disclosure: Generative Artificial Intelligence tools (ChatGPT, Gemini, Perplexity, HeyGen, Napkin AI) were used in the production and translation of the content in this document. AI was utilized solely as a supportive tool: all outputs were validated based on our own sources, subjected to expert review, and modified as necessary. The project implementers maintain full responsibility for the accuracy and originality of the final content.

Open licence
This work is licensed under the Creative Commons Attribution-ShareAlike 4.0 International Licence (CC BY-SA 4.0).

Co-funded by the European Union

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Education and Culture Executive Agency (EACEA). Neither the European Union nor EACEA can be held responsible for them.

Content

1. Introduction to the ECA Teacher's Handbook 4
Aims of the Teacher's Handbook 4
Who is the Handbook for? 4
The logic of the Handbook 4
User Path Map 5
2. From Theory to Practice 6
Curriculum and Handbook 6
You do not have to change everything at once 6
What does the Handbook offer? 6
Mini self-reflection exercise 7
3. Moving from the Traditional Teacher Role towards the Facilitator Role 8
Structure and freedom 9
Conflict management 9
Routines and subroutines 9
Motivation 10
Independence and collaboration 10
The learning space can change too 11
Relationships within the school 11
Change is a process for the teacher 11
Change is a process for the students, too 11
Change is a process for parents, too 11
Mini self-reflection exercise 11
4. Teacher Self-Reflection 13
Self-reflection is part of ECA practice 13
Self-reflection question bank 13
How can we use the question bank? 14
Mini self-reflection exercise 14
5. ECA in Practice 15
From Principles to Practice 15
How is an ECA session structured? 15
Two important areas of ECA practice 16
A traditional lesson turned into an ECA lesson 17
Learning resources 18
Typical difficulties – and possible responses 18
Safety and autonomy 18
Mini self-reflection exercise 19
6. ECA and Science 20
Developing thinking through nature 20
Science education in the Hungarian and Polish national curricula (Grades 1–3) 20
How do we approach science through ECA? 20
Science learning in ECA form 21
From understanding to action – sustainability 22
Beyond science – transferring skills and knowledge 22
Mini self-reflection exercise 23
7. Individual and Differentiated Learning Organisation 24
Learners' needs 24
Mapping individual learning differences 25
Personal touch, relationships, mentoring 25
Experiences of success 26
Tips for individual and differentiated learning organisation 26
Integrating independent and group learning 27
Mini self-reflection exercise 27
8. Monitoring and Assessment 28
Documenting individual progress 28
Assessment and self-assessment 28
Feedback 30
Teacher feedback and formative assessment 31
ÖMini self-reflection exercise 32
9. Adapting and Creating Your Own ECA Materials 33
10. Short lesson plans from practice 34
Example 1: The Eart (LP1) 34
Example 2: Volcanoes (LP3) 35
Example 3: The Human Body (LP6) 35

1.
Introduction to the ECA Teacher's Handbook

Aims of the Teacher's Handbook

The aim of the Teacher's Handbook is to provide practical support for educators in flexibly integrating the ECA approach - emotional development, science and ecological learning, and active engagement - into a traditional framework based on the existing curriculum.

The Handbook supports educators in gradually moving from the traditional teacher role towards that of a facilitator and in developing the competencies required for this role, which differ from those needed for traditional classroom management. It helps them create active, meaningful and collaborative learning situations for students.

It provides support for integrating SEL into subject learning, for differentiation and responding to individual needs, and for giving students greater freedom of movement, more choices and a more active role in their own learning. It promotes experiential and nature-based learning, curiosity, collaboration, engagement and responsibility.

To support this process, the Handbook provides clear and accessible methodological guidance that has been tested repeatedly, frameworks that go beyond conventional classroom settings (e.g. outdoor learning), structured activities and concrete teaching materials that can be used in practice.

These are intended to increase teachers' confidence in applying a person-centred approach. The examples demonstrate how the three elements of ECA can be connected with science and ecological content and sustainability topics, while also providing models for teachers to create their own ECA materials.

The aim is not to adopt a new school system or a specific pedagogical model. ECA can be linked to the existing curriculum and national science requirements, applied in different school and learning environments, and used in a modular way: teachers can begin the process of change with small elements. The Handbook is intended to provide practical support throughout this process, helping teachers not to feel that they have to manage change on their own.

Who is the Handbook for?

The primary target group of the Handbook is teachers working in traditional and alternative schools, particularly those teaching the first three grades of primary school. It is also intended to provide practical support for nature and environmental educators, ecologists, professionals working in camps, forest schools and other out-of-school learning environments, as well as young teachers and student teachers with either science or humanities backgrounds. The approach may also be relevant for special education teachers and for educators working with students with special educational needs, particularly neurodivergent learners.

The logic of the Handbook

1. **Approach and teacher role** → the foundations of ECA, the facilitator role and teacher selfreflection. *(Chapters 1–3)*
2. **ECA in practice** → integrating Emotion–Cognition–Action into teaching, with particular attention to science learning. *(Chapters 4–5)*
3. **Personalisation and monitoring** → differentiated learning organisation, individual learning pathways, self-assessment and formative assessment. *(Chapters 6–7)*
4. **Developing your own ECA practice** → adapting existing materials and creating your own ECA activities. *(Chapter 8)*

The Handbook therefore follows a simple process: first we understand the ECA approach, then we apply it, personalise it and monitor how it works, and finally we become able to create our own ECA materials.



User Path Map

- If you are a beginner teacher: start with the templates and short game descriptions. (Page 34)
- If you are an experienced teacher: look for differentiation tools and integration ideas. (Page 24)
- If you are new to ECA: start with its basic principles and how it works in practice. (Page 6)
- For your own teacher role: see the facilitator role and teacher self-reflection. (Page 8)
- For lesson planning: see the Emotion–Cognition–Action triad. (Page 15)
- For group dynamics, emotions and conflicts: see the Emotion block and communication. (Page 9)
- For knowledge building: see Cognition. (Page 15)
- For science learning: see observation, questioning, direct experience and activities. (Page 20)
- For planning activities: see Action. (Page 16)
- For individual differentiation: see learning pathways, needs and choices. (Page 24)
- For assessment: see self-assessment, formative feedback and monitoring progress. (Page 28)
- For creating your own ECA materials: see adapting materials and developing your own activities. (Page 33)

2.
From Theory to Practice

Curriculum and Handbook

The essence of the ECA (Emotion–Cognition–Action) approach is that emotional and social development, cognitive learning and active engagement should not appear as separate areas in teaching, but should form a learning process in which they reinforce one another. One of its fundamental aims is for the teacher to provide children with the greatest possible learning autonomy within the safest possible learning environment.

The theoretical foundations of the approach are presented in the ECA Curriculum. It introduces the ECA approach and explains the connections between emotional and social learning, science and ecological learning, active engagement and age-appropriate physical activity, as well as the logic of skills development. The Handbook translates these principles into practice by offering tangible frameworks, methodological guidance, concrete examples, structured activities, practical considerations, lesson plans and templates.

Teachers do not have to start from scratch. Many elements of ECA are already present in teachers' everyday practice: attunement, questioning, providing choices, group work, experiential learning, movement, independent discovery and reflection. What is new about ECA is primarily that it brings these elements together within a coherent framework and deliberately connects emotional involvement, cognitive learning and action.

You do not have to change everything at once

ECA can be used in a modular way. There is no need to introduce an entire system overnight. Teachers can start with familiar elements, try out a few ECA activities, or consciously introduce one new element into their existing lessons. Over time, these can gradually develop into more complex ECA learning situations.

What does the Handbook offer?

The Handbook clearly explains this process. It covers:

- methodological guidance on working as a facilitator;
- connecting the Emotion–Cognition–Action elements;
- science and ecological learning;
- differentiation;
- integrating ECA into the existing curriculum.

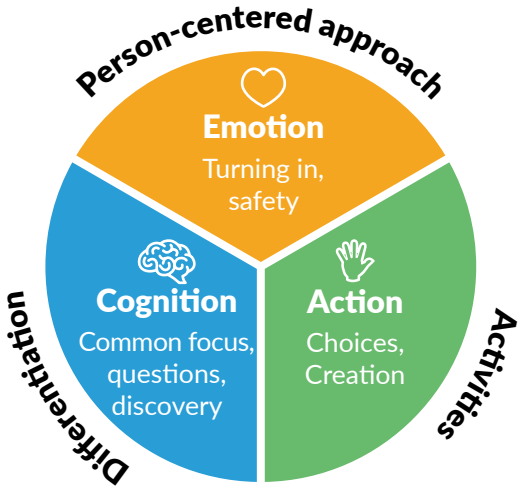
The 50 lesson plans, worksheets, project outlines and homework ideas developed within the ECA project can be used directly, but they also serve as models for creating teachers' own ECA materials.

The examples in the Handbook should therefore not be treated as recipes. An activity can be adapted to another topic, age group, group of learners or learning environment. The ECA approach can be used in both traditional and alternative schools, inside or outside the classroom. The aim is for teachers to select and adapt elements that genuinely work with their own group of learners.

ECA is therefore not another programme added on top of existing teaching. It is a way of organising and connecting elements that are often already present in teaching practice so that Emotion, Cognition and Action form a coherent learning process.



ECA QUICK MAP





Mini self-reflection exercise

- **Which idea from what you have read so far has stayed with you most?** Write it down and add: What does this mean for you in your own classroom?
- **On a scale of 1–10, how much do you currently feel like an "ECA teacher"?** If you would like to move forward, what is the first concrete step you would take?

3.
Moving from the Traditional Teacher Role towards the Facilitator Role

In the ECA approach, the teacher is not simply someone who applies the framework, but its most important carrier. Differentiation, assessment and decisions about the organisation of learning all depend on the teacher's personal presence, sensitivity and professional responsibility. The ECA approach therefore cannot be separated from the person of the teacher: methodological elements alone do not guarantee person-centred practice. Creating safety, the quality of connection with children, the timing of decisions and reflective presence are what bring the framework to life.

Moving towards a facilitator role means that the teacher gives up being the sole director of the learning process and the sole transmitter of knowledge. Instead, the teacher creates learning situations in which children can ask questions, make choices, work independently or together, gain experience and take responsibility for their own learning. The teacher's task increasingly becomes to provide the necessary framework, safety and transparency.

Consideration	Interactive traditional class	ECA class
Motivation	Grade	Unique experience,
Assessment	Grading based on performance	Process-oriented self-assessment
Differentiation	Difficult to solve	Naturally comes
Action	Limited	Multiple, experiential, movements
Cognitive processing	Frontal, based on instructions	Self-contained, exploring, individual
Emotional involvement	Low-medium	High
Role of the learner	Student, receptive	Active, experimental
Focus	Transfer of knowledge	Experiences, complex development
Mode of operation	Instructions, problem solving	Unique products, complex

The table illustrates the direction of change: from frontal, instruction-based cognitive processing towards independent and exploratory learning; from the learner as listener and recipient towards an active, experimenting role; from limited action towards varied, experiential and movement-based activities; and from performance-based grading towards process-oriented self-assessment.

Structure and freedom

Structure is an important element of person-centred practice. If children are to have greater autonomy, they need stable routines, transparent frameworks and genuine points of choice. The teacher briefly frames what is going to happen, why it is important and where there is room for choice. Children can regularly encounter genuine choices during the day – for example regarding the task, tools, pace or role.

In action-based learning, learning also takes place through movement, creation and experimentation, rather than exclusively through verbal or frontal teaching. This requires flexible use of time: the teacher observes the rhythm of the group and, when necessary, slows down, pauses or changes direction. If a child seems unable to find their place, the teacher asks, shows interest and, with the child's permission, offers support.

This way of working is both structured and flexible. Structure provides safety, while flexibility allows the teacher to respond to different learning paces, the current state of the group and individual children's signals. The teacher is therefore present in the learning process in a different way: less as a director determining every step and more as a professional who frames, observes and supports the learning process.

Conflict management

Conflict management is a particularly concrete example of how the teacher's role changes. The teacher does not decide who was at fault and what the consequence should be, but facilitates a process in which children understand the situation, express their feelings and needs, and look for a solution together. This is closely connected to active listening, autonomy, responsibility and personcentred teaching practice.

The concrete practice of conflict management is discussed in Chapter 5.

Routines and subroutines

Routines make our lives easier by allowing us to focus our attention on important and changing tasks and circumstances. At the same time, routines can also limit us: habit may restrict creativity. What kinds of routines can support teachers in implementing ECA? The following list is not exhaustive, but offers some concrete examples:

- **Arrival and attunement:** Begin the day with a short, structured but freely accessible activity – a circle, movement, silence or a chosen activity – signalling that every child may arrive in their current state. The Cat Cards, developed by the Rogers Foundation to express emotions, can also be used for this purpose.
- **Opportunity to indicate one's state:** Children can show how they are feeling in a simple way – through a colour, gesture, word or card – without having to explain themselves.
- **Building in points of choice:** Genuine choices regularly appear during the day concerning tasks, tools, pace or roles.
- **Short shared framing:** The teacher explains what will happen, why it matters and where there is room for choice, thereby creating safety and transparency. The greater the space for meaningful choice, the better.
- **Action-based learning:** Learning also takes place through movement, creation and experimentation, rather than exclusively through verbal or frontal teaching.
- **Flexible use of time:** The teacher observes the rhythm of the group and, when necessary, slows down, pauses or changes direction rather than rigidly following the original plan. If a child seems unable to find their place, the teacher asks, shows interest and, with the child's permission, offers support.
- **Observation instead of assessment:** During the day, greater emphasis is placed on observation and reflection than on judging or correcting.
- **Treating conflicts as learning situations:** Conflicts that arise are not seen as disruptive factors, but as situations that can be explored and used for shared learning.

- **Closing and looking back:** At the end of the day, a short reflection: What was easy? What was difficult? What felt good? What went well? What should we change?
- **Predictable, human presence:** The teacher's consistent, authentic and emotionally available presence accompanies the children throughout the day.

Motivation

Every chapter and idea in the Teacher's Handbook is based on the assumption that natural learning conditions are motivating in themselves. The Handbook was created so that each teacher can put into practice those elements that they can use authentically.

The following list contains elements of a complex system. This system is person-centredness, which provides the foundation for the ECA approach. We do not claim that all its components are included here, but the list is already quite extensive:

- Taking individual developmental pace into account.
- Actively involving learners in the learning process and adopting a facilitator role.
- Creative, experiential activities.
- A differentiated approach.
- Supporting community activities.
- Creating an atmosphere of trust through the teacher's attitude.
- Encouraging independence and responsibility through conscious pedagogical practice.
- Valuing learners' personal experiences.
- A positive teacher attitude and supportive presence.
- Recognising the individuality of learners.
- Movement-based and creative activities.
- Creating an environment in which it is safe to make mistakes.
- The enjoyment of creative and practical activities.
- Building on learners' interests.
- Noticing and acknowledging learners' successes and efforts.

Independence and collaboration

Independent and group learning are not alternatives to one another. Learners can work independently or in pairs at their own pace and then share the information they have gathered with the group, where joint discussion, forming opinions and decision-making can follow. Clear roles within the group – for example, coordinator, note-taker or timekeeper – support organised work and individual responsibility. The teacher can also support children's time management by providing clear time frames.

Learner autonomy therefore does not mean leaving learners on their own. One of the important tasks of the facilitator is to create the conditions for collaboration. This can be supported by rules developed jointly by the learners, feedback after group work, the conscious use and rotation of roles, and a supportive, empathic teacher–learner relationship.

We have also collected some points that can help teachers prepare to build more personal relationships with learners and take on an effective mentoring role.

- Take time to really listen to the child, paying attention to non-verbal signals as well.
- Have regular individual conversations with the child about personal matters and feelings.
- Consciously identify and reinforce each child's individual strengths.
- Respect confidentiality and the rules of trust.
- Develop individual goals with each child and observe changes together.
- Be consistent and authentic, and lead by example.
- Encourage experimentation and independence.
- Accept individual circumstances.
- Organise shared activities.
- Give regular, constructive feedback.
- Active listening, empathy, supportive and a non-judgemental attitude are all important.
- Practise clear self-expression, honesty, flexibility and adaptability.

What else can the teacher do for learners who find change more difficult than others, for example autistic children?

- Always announce changes in advance and explain why they are useful or necessary. Give learners time to prepare.
- Whenever possible, begin with small changes before introducing larger ones.

- Frame new situations positively: “This will be different from what we are used to...”
- Avoid using the word “must”. Make sure you yourself are convinced that the change is worthwhile.
- Reinforce successful adaptation: “Thank you for adapting so well!”
- Inform parents about the change.

The learning space can change too

Moving beyond traditional classroom arrangements does not necessarily mean leaving the classroom. Change can begin simply by adapting the physical environment to the learning situation: not always working in the same seating arrangement, in the same place or in the same format.

The Handbook emphasises creating a safe and engaging physical learning environment and continuously looking for alternatives to working at desks. Flexible use of space allows the same learning process to include individual work, pair or group collaboration, movement, creation and shared discussion. Learning outside the classroom is discussed in more detail in the ECA + Science chapter.

Relationships within the school

ECA works more effectively when the teacher is not alone. Colleagues can share experiences, think together and form teacher teams. Cooperation with school leadership is also important, particularly when the ECA approach extends beyond the practice of a single class.

Relationships with parents should be based on trust, clear boundaries and regular, transparent communication. Parents should be involved in ways that also protect the child’s personality and confidential disclosures. In every direction, cooperation should support two-way sharing of experience and the child’s development.

Change is a process for the teacher

The facilitator role does not develop overnight. It is worth moving towards it gradually.

1. Emotion – introducing routines

1. Short 5–10 minute morning starters using emojis.
2. Simple conflict-management games.
3. Keep a short teacher journal.

2. Introducing ECA lesson plans

1. One or two ECA lessons per week using familiar topics.
2. Give children choices.
3. Use a scale for assessment at the end of the lesson.

3. Deepening differentiation

1. Get to know children’s learning habits and preferences.
2. Initially offer 2–3 choices in the Action section.
3. Introduce mini learning contracts with 3–5 children.

4. Full integration

1. Four or five ECA lessons each week.
2. Share your experiences with two new colleagues.
3. Reflect on five self-reflection questions each week.

The transition can also be approached through the level of challenge:

Weeks 1–2 – easy routines

1. A daily five-minute check-in circle.
2. Two-minute movement breaks during lessons – jumping, stretching.
3. Using open-ended questions.

Weeks 2–4 – moderate challenge

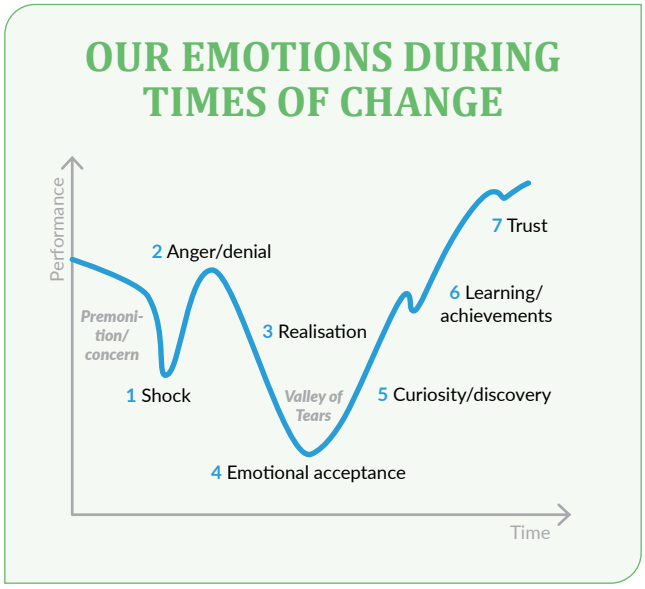
1. Groups of three or four, with changing group composition.
2. Offer 2–4 choices.
3. Introduce a restorative circle: “What happened?” – “What can we do?”

Months 1–3 – more challenging steps

1. Plan an ECA lesson.
2. Integrate the curriculum flexibly.
3. Design a complete learning unit.

Person-centred practice requires competencies that differ from traditional classroom management. Teachers need to learn how to provide structure while also allowing space; attend to individual children and the group as a whole; respond to different paces; support collaboration; and, when necessary, change the original plan. Clear methodological guidance and structured activities help teachers themselves feel secure during this process.

Understanding the change curve can provide another useful tool. It can help teachers interpret children's reactions more realistically when introducing a new situation, rule or form of learning. The model associated with Elisabeth Kübler-Ross illustrates how change may initially evoke uncertainty, resistance or even regression before adaptation and acceptance emerge. Recognising where a child or group may be in this process can help the teacher respond with greater patience rather than interpreting difficulties as personal failure. It helps to see change as a natural developmental path.



The source: https://www.icgstadium.hu/insights/hogy-eljuk-meg-a-valtozast/

Change is a process for the students, too

For children who find changes in familiar routines particularly difficult, it is especially important to explain in advance what will happen and why, allow time for adjustment, and proceed in small steps whenever possible. Predictable frameworks and talking about the change can help transform something new from a source of uncertainty into a gradually manageable new situation.

Change is a process for parents, too

The following points can help maintain a constructive relationship:

- Maintain clear, public communication channels – letters, parent forums, information evenings – with clearly defined boundaries.
- Establish shared school principles and rules accepted by both teachers and parents, including procedures for confidentiality and conflict intervention.
- Give parents trust-building but clear and consistent feedback so that they can see that the school acts in the child's interest while protecting confidential information.
- Make clear what and how we want to share with parents as partners, and which decisions remain the professional responsibilities of the teacher.
- Ask parents to be open to dialogue and develop communication rules together; share information about the child that may be important for the teacher; stay involved and informed; and respect professional confidentiality, maintaining trust even in difficult situations.

The aim of facilitative practice is therefore to create a learning environment in which safety and autonomy can grow together.

Mini self-reflection exercise

- **Think of a situation in which you managed to step back from the traditional teacher role** and accompany the children more as a facilitator. What helped you resist giving them the solution?

- **Think of a change you have recently introduced in your class.** Consider how the children, you yourself and the parents responded to it. How was it received, and how did the experience feel for you?

4.
Teacher Self-Reflection

Chapter 3 shows how the teacher's role changes; this chapter focuses on how teachers can continuously examine and develop their own practice.

In ECA, self-reflection is not a separate technique but an approach that runs through the whole of pedagogical practice. It helps teachers become aware of their own attitudes, decisions, emotional reactions and methodological choices, and then use their experiences to further develop their practice.

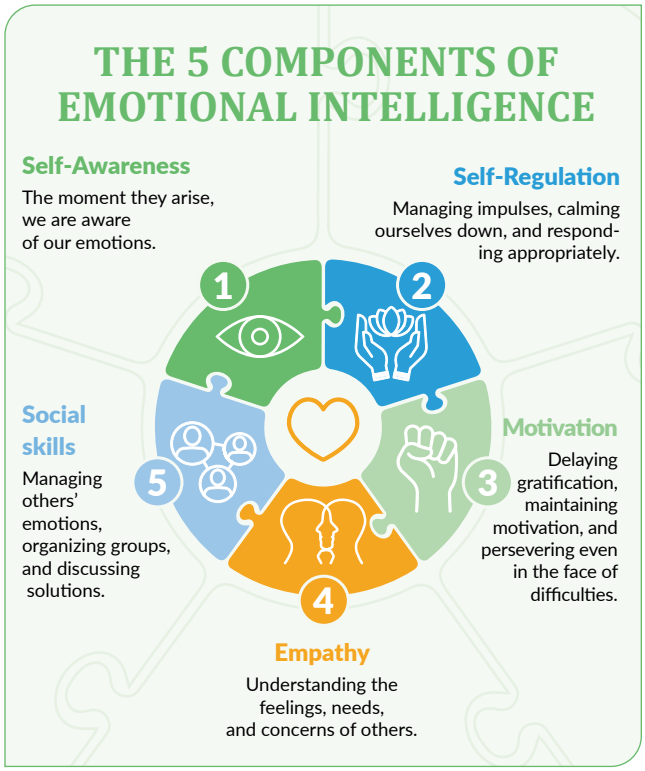
If we consider self-reflection important for children, it is worth expecting the same from ourselves. Teachers have more experience and professional knowledge, but understanding their own emotional and social functioning is also an ongoing learning process. Teacher self-reflection is not very different from the way we think about children's emotional and social development. This is no coincidence. We may know more on the cognitive side, but in the emotional domain there is always just as much for us to learn as there is for young children – only at a different level. Let us be our own teachers and learn from the children.

Emotional intelligence plays an important role in this: recognising and regulating our own emotions, empathy, managing our relationships, and understanding our own motivation are important not only for children's development, but also for becoming more aware of our own practice as teachers.

Self-reflection is part of ECA practice

The Curriculum presents in detail the areas of teacher practice connected to the ECA approach that are worth considering in self-reflection. Rather than explaining them again here, we turn them into practical self-reflection questions.

The following question bank moves through three interconnected levels: short, even daily reflection → deeper examination of our own attitudes and practice → conscious professional development goals and change.



Self-reflection question bank

Connection, safety and conflict

- Do the children feel safe? Do they ask questions, speak up and take initiative?
- How do I react when a child makes a mistake or does not understand a task?
- Is there a child I find more difficult to connect with? What might be the reason?
- Was I able to see a conflict as an opportunity for learning?

Autonomy and participation

- When did I provide a genuine choice?
- What were the children actually able to decide?
- When did I step back and allow them to look for their own solution?

Emotion – Cognition – Action

- What emotional state was I able to support?

- What kind of thinking process did the topic initiate?
- Was there genuine action, direct experience or creation?

Individual differences and differentiation

- Which child might have benefited from a different tool or a different type of task?
- Is there a child whom I tend to compare with others?
- For whom would I like to introduce one small, personalised change in the coming period?

Communication and feedback

- What can I learn from the feedback children give me?
- From whom would it be useful to ask for external feedback now?
- How do I regularly collect feedback from children and others about my own practice?

Process, outcomes and assessment

- Did I pay attention to the learning process as well as the outcome?
- Did my feedback help the child understand what they had learned and how they had developed?
- Did assessment support further learning, or did it mainly judge performance?

My own teaching practice and development

- Which teaching role do I feel most comfortable in?
- What values do I represent as a teacher? How are these reflected in my practice?
- In what areas would I like to develop?

How can we use the question bank?

This question bank is not a checklist, and there is no need to work through it in order. From time to time – for example, once a month – choose one or two questions that resonate with you or feel particularly relevant at that moment. Formulate your own questions as well. Later, it can be useful to return to a question you considered earlier and see whether anything has changed. Reflection can take the form of a few minutes of thinking, a short journal entry, a conversation with a colleague, or even documenting or video-recording a lesson.

In self-reflection, the things that are working well are just as important as those we would like to change. The aim is not to judge ourselves, but to recognise where we are now and what our next step might be. Notice what is already working well, learn from your experiences, and be willing to make changes. The key is to develop the habit of regular reflection and consciously gathering feedback.

Mini self-reflection exercise

- **Choose one teacher role** – for example, facilitator, mentor or team player – and mark on a scale from 1 to 10 how comfortable you currently feel in that role.

- **Write down one small situation in which you will consciously practise this role during the coming week,** for example: "During this group activity, I will only ask questions instead of providing the solution."

5.
ECA in Practice

From Principles to Practice

The ECA programme places children's emotional involvement, active participation and learning through their own experience at the centre: their own pace, motivation and connection to their needs. Each child is approached individually, taking age-related characteristics into account.

The three areas of ECA do not mean that exactly the same amount of time must be devoted to all three in every situation. The Emotion section, for example, deliberately receives more space in this Handbook because this is where teachers are likely to encounter the most new elements. This does not mean that Emotion should have a greater weight than the other two areas.

In ECA, therefore, it is important to notice not only when one area is missing, but also when one becomes persistently dominant. The aim is for emotion, understanding and action to support one another throughout the learning process.

How is an ECA session structured?



Emotion – the emotional dimension

1. **Connection:** a short moment of attunement – making eye contact, a short game, etc.
2. **Reading the emotional state:** a quick indication of how ready each child is for learning. Simple tools can help children recognise and name their emotions. If necessary, brief emotional support or framing can be provided.
3. **Moving towards Cognition:** involving the children, creating personal relevance, awakening curiosity and preparing the ground for the topic.

The purpose of the Emotion block is to create an emotionally safe atmosphere and establish a connection with one another and with the learning content. This opens children's attention and supports the lasting integration of information. Curiosity is a powerful catalyst, while opportunities for choice and for making mistakes are essential to autonomy.



Cognition – understanding and thinking

1. **Defining a shared focus:** a short, clear goal and topic – one question or one problem. The cognitive content may come from the curriculum or from a topic raised by the children.
2. **Exploration:** through their own questions and observations, at their own pace and by expressing their own ideas, learners develop their relationship to new knowledge.

3. **Organising together:** discovering key concepts, rules and connections, drawing conclusions and organising what has been learned. Alongside knowledge, the aim is to develop skills and critical and creative thinking.
4. **Brief consolidation:** a short, experience-based summary or visual representation that helps children become aware of their new knowledge.

The cognitive dimension of ECA is essentially the familiar content of a traditional lesson: acquiring knowledge and developing skills. Teachers already know this area well. ECA does not provide more or less Cognition.



Action – activity and experience

1. **Moving into action:** offering an activity. The aim is to deepen knowledge through different pathways.
2. **Creating or practising:** carrying out the activity. The aim is to gain experience, practise and experiment, and to connect understanding with action.
3. **Sharing:** a brief presentation. The aim is to reinforce learning and provide a response from the community.
4. **Reflection:** "What happened inside me, and what did I learn?" Children look back on their own experience and connect emotion, thinking and action. The aim is to consolidate the learning experience.

Action is the experiential stage of the ECA process in which learning turns into activity. This is where what was initiated during the emotional and cognitive preparation becomes visible: children show themselves through what they do. The teacher's role at this stage is to maintain a balance between a safe framework and freedom. In this way, learning can become a genuine experience that comes from within. The Action dimension itself is not new to teachers, but ECA increases opportunities for choice and, with them, children's decision-making and responsibility. Activities can take many forms: real-life problems, communication tasks, artistic activities, digital tasks, sensory and movement-based exercises, making and building, environmental activities, visits to external locations, projects and thematic days.

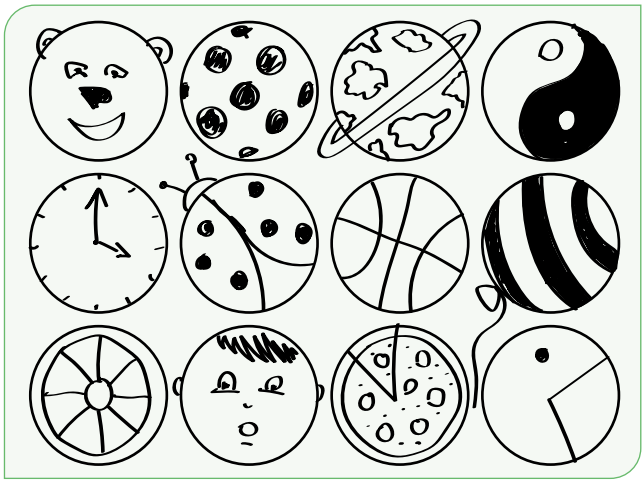
Two important areas of ECA practice

Among the many practical applications of ECA, we highlight two examples that are also emphasised in the Curriculum: creativity, which supports independent and creative functioning, and communication and conflict management, which support relationships.

Creativity

Following Rogers' thinking, creativity has a prominent place in our Curriculum. It is therefore worth exploring it here in more detail, even though it often emerges spontaneously and can be connected to all three ECA areas. At the same time, creativity can also be supported through deliberately planned activities. Children's spontaneous ideas always deserve careful consideration and are often more important than our original plans.

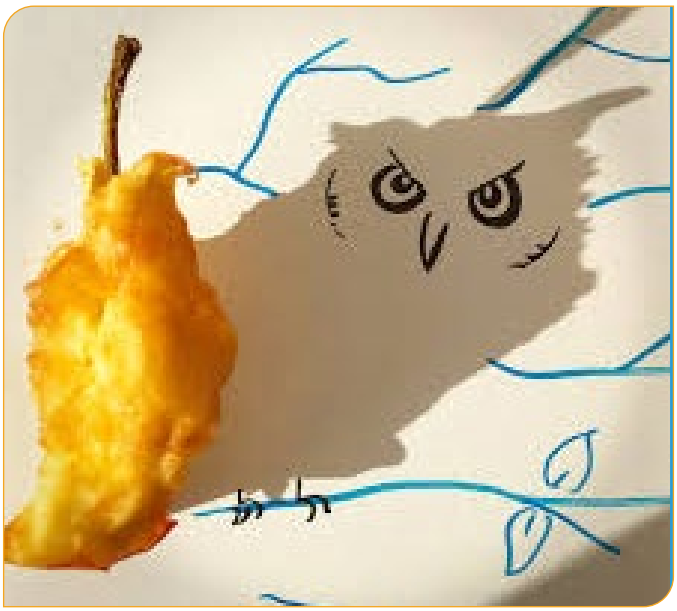
The Torrance Test



One useful tool for exploring creativity is the Torrance Test. Its tasks make it possible to observe,

among other things, the number, variety, originality and elaboration of ideas, and can provide insight into divergent thinking.

Creativity across subjects



Creativity can have a place in ECA as a separate area of skills development, but it can just as naturally be integrated into different subjects. For example:

- **Writing:** Rewrite stories with a new ending or new characters.
- **Reading:** Role-play based on stories the children have read.
- **Mathematics:** Use patterns from nature for mathematical tasks. Find five square shapes in the garden.
- **Environmental studies:** Make musical instruments from natural materials.
- **Music:** Invent melodies and rhythms using different natural materials.
- **Movement:** Free movement-based storytelling, dance and dramatisation based on stories..

Communication

Communication in ECA is not a separate technique. Recognising and expressing emotions, understanding another person's perspective, expressing one's own needs and looking for shared solutions are all part of person-centred practice.

It is useful to practise different communication situations with children: exploring answers to open-ended questions, observing communication mistakes, analysing everyday situations, playing empathy-building games, or identifying and describing the emotions of people shown in pictures.

Conflict as a learning situation

In a person-centred environment, conflict is not a "disruptive factor" but a learning situation. Children learn that tension can be managed and relationships can be preserved. Here too, the teacher acts as a model: rather than looking for someone to blame, the teacher helps both parties express what they need.

Concrete steps in mediation

1. **Stop the situation:** "Let's stop for a moment. What happened here is important."
2. **Listen to both parties separately, without judgement:** "Tell me what happened from your point of view."
3. **Name emotions:** "How did that feel for you? Did you feel embarrassed...?"
4. **Express needs and requests:** "What would you like to happen now? What are you asking from the other person?"
5. **Look for a shared solution:** "What solution could work for both of you?"

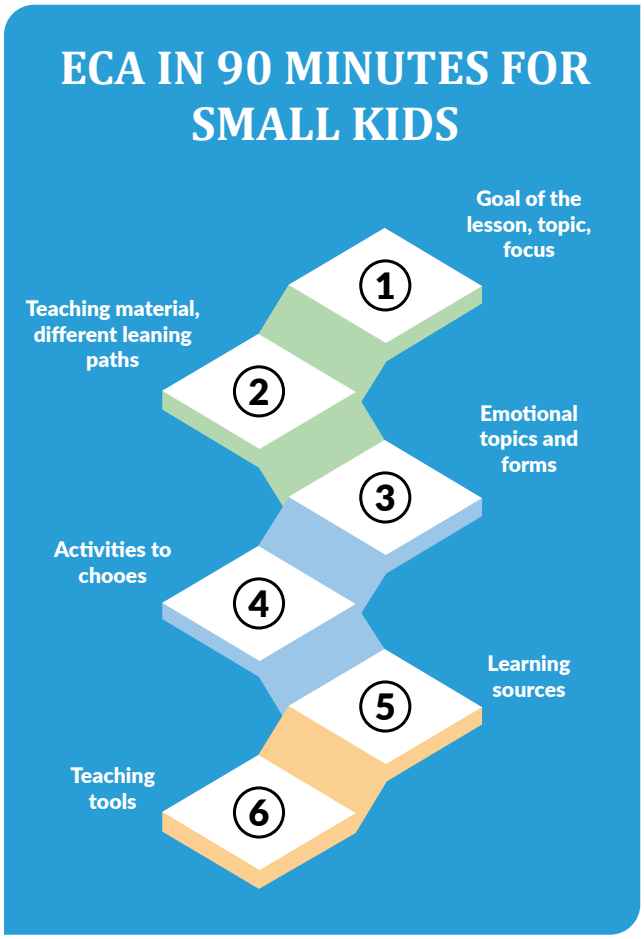
Sentence starters for children

- "When ... happened, I felt ..."
- "Next time, I would like ..."
- "I'm sorry that ...; next time I'll try to ..."
- "Right now, I need ..."

A traditional lesson turned into an ECA lesson

The first steps in preparing an ECA lesson are not new to teachers. The main difference is that we think more consciously about emotional engagement, offer a wider range of activities to choose from, pay attention to the diversity of learning resources, and include reflection at the end of the lesson.

Steps in the preparation process – a planning guide



1. What is the topic? Cognitive topic, goals
2. Cognition: Topic connected to prior knowledge
3. Emotion: relevant situation and personal connections to the topic
4. Action: Choices for activities. Instructions.
5. Learning resources
6. Forms of production
7. Learning tools
8. Reflexions

Learning resources

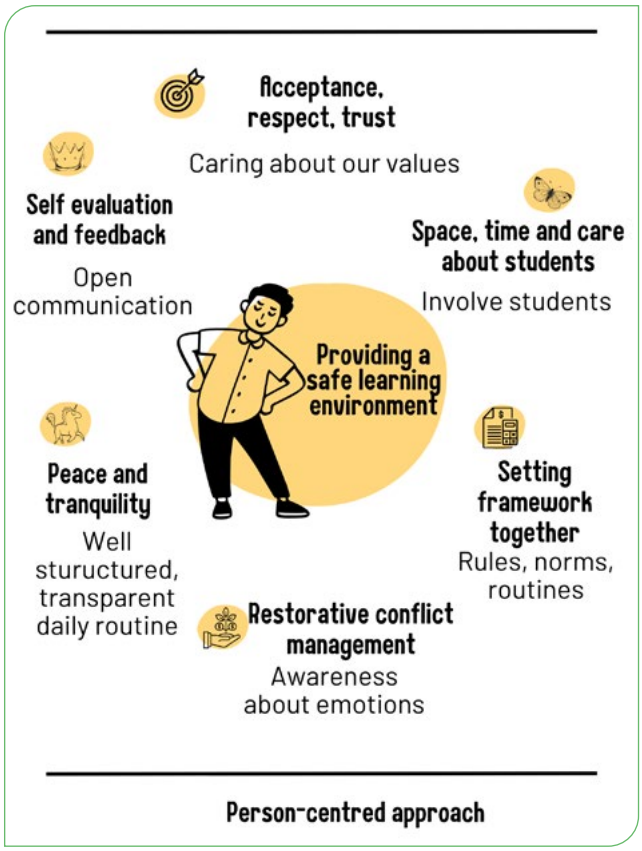
Textbooks, workbooks, wall maps, aids, graphic and visual tools on the one hand, and interactive whiteboards, tutorials, e-textbooks, multimedia educational packages, smartphones, tablets and laptops on the other. This duality characterizes our time. A tudatos, kombinált használat változatosabbá teheti a tanulást.

Typical difficulties – and possible responses

Several recurring difficulties may arise when introducing ECA. It is useful to look at them consciously.

- **Lack of time:** occasionally plan longer learning periods by combining topics.
- **Learner resistance:** rely on group rules created together.
- **"It will become chaotic":** clear frameworks and short, simple instructions help.
- **Not everyone dares to speak:** offer participation in pairs, small groups or without words.
- **Maintaining quality:** if children try to avoid the work, give precise feedback. If the problem persists, explore the causes and possible solutions together with them.
- **Lack of a supportive environment:** an individual teacher can still begin working with ECA.
- **Managing emotional challenges:** conscious preparation helps.
- **Pandora's box:** talking about emotions may bring serious problems to the surface. This is a genuine concern. Therapeutic exploration is not the teacher's task. Seek appropriate professional support when necessary.
- **Technical or spatial limitations:** ECA can be started even in a simple classroom.
- **Maintaining long-term motivation:** moving away from established daily routines takes time.
- **Difficulties with assessment and feedback:** process-oriented assessment requires a different kind of preparation.







- **Fear of mistakes:** make it clear that mistakes are a natural part of the process.

Safety and autonomy

One of the fundamental aims of ECA is for safety and autonomy to be present at the same time in the learning process. The following two figures summarise the kinds of teacher decisions and ways of working that can help create these two conditions. Through ECA, we become more conscious of seeing the whole learner behind the learning content.

Mini self-reflection exercise

- **Think back to a lesson in which the Emotion–Cognition–Action triad was already present,** perhaps without you consciously planning it. What happened at the beginning, in the middle and at the end?
- **Choose an upcoming topic and describe in three sentences what emotional state,** thinking process and action you would like to see.

6.
ECA and Science

Applying a person-centred approach to methods of scientific inquiry, classroom practice and skills development

Developing thinking through nature

Nature is not only a subject of learning, but also a medium for thinking, discovery and personal development. We learn to think with the help of nature and through nature. Our approach is based on four complementary dimensions:

- **Scientific attitudes** – developing curiosity, openness, respect for nature and evidence-based thinking.
- **Methods of inquiry** – observation, questioning, investigation, measurement, comparison and simple experimentation.
- **Pedagogical practices** – creating learning situations based on direct experience, discovery and reflection.
- **Science skills** – consciously developing observation, description, comparison, classification, measurement and inference.

In the following sections, we show how these dimensions connect to the aims of the Hungarian National Core Curriculum for Grades 1–3.

Science education in the Hungarian and Polish national curricula (Grades 1–3)

In both countries the National Core Curriculum gradually introduces scientific thinking and sustainability during the first years of primary education. The emphasis is not on transmitting factual knowledge, but on gaining experience, scientific inquiry, and developing curiosity, respect and responsibility towards nature. ECA builds on this foundation by complementing cognitive learning with personal and emotional connection and real activity. Let us first look at the cognitive dimension.

How do we approach science through ECA?

Skills and questions

Questions do not have to arise only from the curriculum. They can also emerge from what surrounds children, what they experience and what arouses their curiosity.

Observation, description and comparison are fundamental thinking skills. Through them, children gradually explore and understand the world around them. These skills develop from sensory experience and wonder towards more systematic and analytical thinking. The ECA approach supports this process through curiosity, conscious observation and children's own discoveries.

Grades 1–2 – Observation

Children explore the world primarily through their senses: What is the world around us like? What is it made of? What is it for? How beautiful it is!

Grades 2–3 – Observation

Attention gradually turns towards understanding relationships, structures and functions:

What is this part of? What structures can I discover? How does it work? What does it do? Why?

Grade 3 – Description

Learners describe the characteristics of objects and phenomena, as well as changes, with increasing precision: What makes it change? How does it change? What does it change from and into? What was it like before? What is it like afterwards?

With sufficient practice, children learn to create vivid descriptions, clearly describe processes and adapt their descriptions to different situations.

Grades 3–4 – Comparison

Learners recognise similarities and differences and organise their experiences. Key concepts include: characteristics; living and non-living; nature and society; same – similar – different; interaction.

Science learning in ECA form

The science learning process follows a simple path:

- **Relevant questions:** We start from the world around us and from real problems.
- **Observation:** Investigation, data collection, description, measurement, comparison and deeper questions.
- **Scientific thinking:** Questions, hypotheses, conclusions and openness to results.
- **Direct experience:** Experimentation, discovery, working with real materials and first-hand experience.
- **Action and activities:** Using knowledge in practice and paying attention to the environment.
- **Transfer to other areas:** Applying these skills in social situations and integrating them into everyday life.

This is not a compulsory lesson structure. It illustrates how science learning within ECA can become a coherent experiential process.

Science content and related skills belong primarily to the cognitive dimension. The learning process becomes an ECA activity when personal and emotional connection and genuine activity are added.

ECA does not add another body of content to be learned. Instead, it helps make existing content experiential, personally relevant and action-oriented.

The life of a tree

If there are several types of intelligence, le tus see what it means in a concrete topic.

1. **Bodily–kinaesthetic intelligence:** Children act out the life of a tree through movement: seed, sprout, growth, storm, falling leaves. Through their bodies, they experience change and adaptation.
2. **Naturalistic intelligence:** They observe a real tree – its leaves, bark, surroundings and inhabitants – and formulate questions about how it coexists with other living organisms.
3. **Logical–mathematical intelligence:** They estimate the age of the tree, count growth rings on a cross-section, and compare the size, shade and water requirements of different trees.
4. **Verbal–linguistic intelligence:** They write a story or short text "from the tree's point of view": what it sees, experiences, fears and enjoys.
5. **Visual–spatial intelligence:** They draw the visible and "invisible" parts of the tree – roots and the underground world – or create a map of its surroundings.
6. **Musical intelligence:** They represent the life of the tree through sounds and rhythms: wind in the leaves, rain, birds and the changing seasons.
7. **Interpersonal intelligence:** In small groups, they discuss what a tree "needs" in order to live and draw parallels with how a community functions.
8. **Intrapersonal intelligence:** Individual reflection: "When do I feel as stable as a tree?" or "What helps me grow?"
9. **Practical intelligence:** Planting a tree, composting, mulching or creating a tree-care plan.
10. **Existential intelligence:** Discussing what time means in the life of a tree, what remains after it, what its existence means and how humans are connected to the living world.



Within this single topic, all three ECA dimensions can be present together: Emotion – a personal connection to the tree and the living world; Cognition – observation, questioning, measurement, comparison and looking for connections; and Action – direct investigation, creation and concrete action for the environment. The tree therefore becomes more than simply a topic to be learned: it becomes a real starting point for understanding the world. This comp-

lex ECA approach can be applied to many different areas of learning.

Direct contact with nature

One particular opportunity offered by science learning in ECA – as with other learning content – is that the setting and tools for learning can easily extend beyond the textbook and the classroom. This does not necessarily require a special programme. We can think in three simple directions.

1. **We go out into nature.**
   The schoolyard, garden, nearby park, stream bank or forest can itself become a learning environment. We can go outside with just one concrete question: What can we find under the tree? Where is the soil wetter? What living organisms can we find in sunny and shaded places?
   By returning to the same place several times, we can also observe changes over time. The important point is that children encounter not only information about nature, but the phenomenon itself.
2. **We bring nature into the classroom.**
   If we cannot go outside, the object of investigation can come to us. Leaves, fruits, stones, soil samples, seeds, plants and other natural materials can be used not simply as illustrations.
   Children can handle, classify, compare and measure them and formulate questions about them. We can also start longer processes so that change itself becomes observable. Many elements of this are already present in everyday classroom practice, such as caring for plants, germinating seeds or animal-assisted education.
3. **We work with real materials instead of only talking about them.**
   This is the shared principle behind the first two approaches. Whenever possible, we observe, touch, measure, compare and experiment with things, and follow how they change over time. For example, the question "What is soil like?" can lead to comparing different soil samples; "How much water do we use?" can lead to measurement; and an environmental problem can lead to shared investigation and action.

The three approaches can naturally be combined. We can go outside to observe something, bring back a sample for further investigation, and later return to the site to check our answers or observe change. In this way, nature becomes not an illustration of the curriculum, but one of the sources of learning.

From understanding to action – sustainability

In ECA, science learning does not necessarily end when a phenomenon has been understood. Action creates opportunities for knowledge to lead to activity: saving water or energy, protecting the local environment or taking part in community action. The point is not to turn every science lesson into an environmental campaign. Rather, when what children have learned can lead to meaningful, age-appropriate action, we should make room for it. Ecological awareness days such as World Water Day or Earth Day, for example, can provide good starting points for complex learning activities.

Beyond science – transferring skills and knowledge

Real-life problems are not organised according to school subjects. Exploring a natural phenomenon can easily connect with mathematics, language, visual arts, technology or social and emotional learning. ECA adds its own three questions: What do I understand? How does it relate to me? What can I do with it? Everyday life also presents many problems in which scientific skills can provide a useful basis for finding solutions.

Science as a school of thinking

Science does more than teach us about nature. It can also develop habits of thinking that children can use in other subjects, social situations and everyday life. Once again, questioning is the starting point. It is worth asking questions that matter to us. Four examples illustrate how thinking learned through science can be transferred elsewhere:

- **Distinguishing fact from interpretation.** Careful observation teaches us first to describe what we actually see or experience and only then to interpret it. This skill can also be used in social situations. In a conflict, for examp-

le, saying "You deliberately disturbed me" is very different from describing precisely what happened. Gordon's three-part confrontive I-message similarly begins with an objective description of behaviour.

- **Evidence-based thinking.** We observe, measure, compare and collect data, and then try to draw conclusions from them. The important question is not only "What do I think?" but also "What is my claim based on?" This is a useful habit in any discussion.
- **Accepting uncertainty and revising our own ideas.** The result of an investigation is not always what we expected. When examining a scientific hypothesis, there may be several possible explanations. We can be wrong, try again and revise our previous ideas in the light of new experience. This ability is just as valuable in everyday disagreements as it is fundamental to scientific thinking.
- **Systems thinking.** In nature, things rarely happen in isolation. We look for relationships, interactions and consequences: what affects what, and how does a system change when one of its elements changes?

These four areas are only examples. Science learning can develop many other thinking skills and attitudes that can be used in other subjects and situations in life. The same content can also be explored in different ways by taking different areas of intelligence into account. A single topic can connect science learning with all three ECA dimensions and with the thinking skills described above. The same reality can be approached from many perspectives: it can be observed, measured and compared; it raises questions; it reveals relationships and changes; and at the same time, we can have a personal and emotional relationship with it.

This chapter has presented the main practical directions for approaching science through ECA. They are not compulsory recipes, but starting points that teachers can adapt to their own curriculum, environment and group of children.

Mini self-reflection exercise

- **Think of a science topic you taught recently.** Where could the children have had more opportunities to observe, ask questions, gain direct experience or take action?

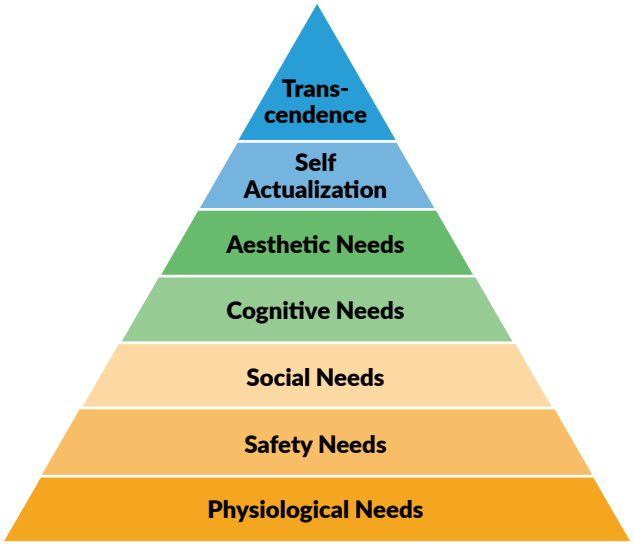
- **Choose an upcoming science topic.** What do you think might be the first question about it that would most interest the children? Then ask them – and compare their answers with your own.

7.
Individual and Differentiated Learning Organisation

We have reached one of the key points of our approach. One of the – ACHIEVABLE! – aims of person-centred education is to create an appropriate, motivating and developmental learning environment for every learner within a group. We are different. Accepting this is an important insight that fundamentally shapes the teacher's attitude.

Learners' needs

Maslow's hierarchy of needs



Maslow's hierarchy reminds us that the state of a child's basic physical, safety and relational needs can significantly influence their ability to concentrate on and participate in learning.

As teachers, we should consider which level requires immediate attention: rest, safety, acceptance or challenge.

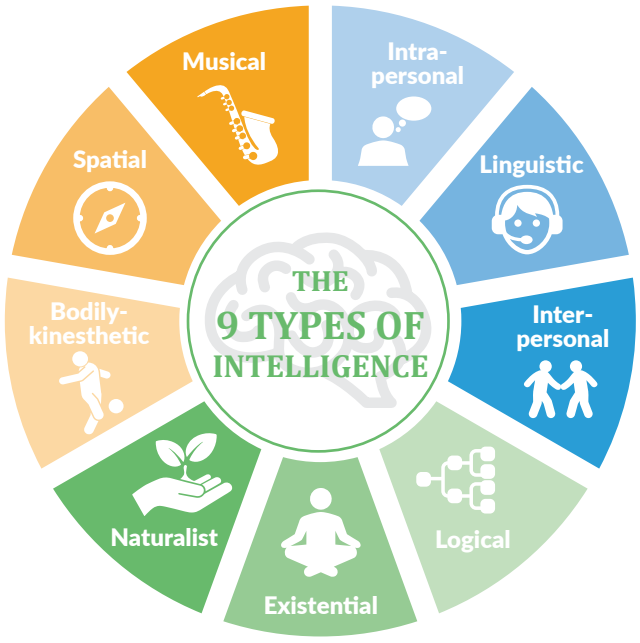
Here is one activity for exploring this question:

Learning Fuel Station: Draw four "fuel pumps" on the board, using words or pictograms:

1. food / drink / rest
2. safety
3. friends
4. challenge

Children place a sticker next to the areas where they currently feel "empty" – no more than two. In a short class discussion, consider together what could be done that day to make learning feel better.

The multiple intelligences perspective

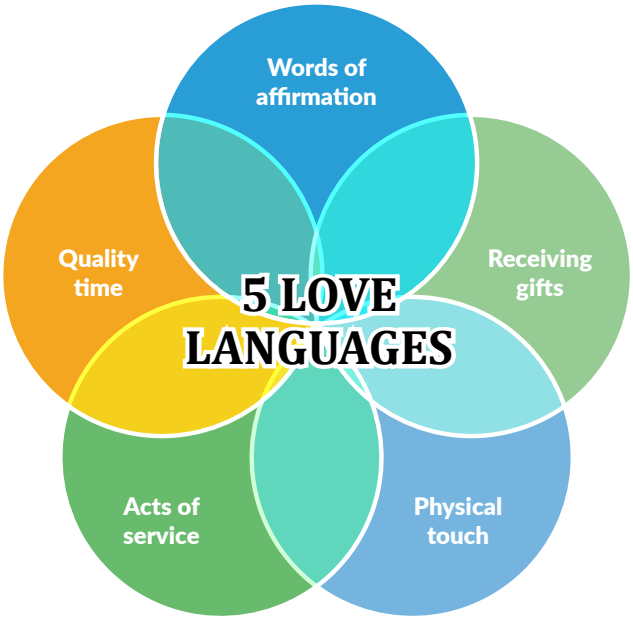


Howard Gardner's theory of multiple intelligences draws attention to the fact that children may show strengths in different areas and may connect with the same topic in different ways. This does not mean placing learners into fixed "intelligence types".

As teachers, we aim to make it possible to approach the same topic through different kinds of activities: for example, linguistic, logical, bodily-kinaesthetic, musical, visual or interpersonal activities. The examples in the list below can help us offer different ways of engaging with a topic.

The key is to offer at least 2–3 different channels within an ECA unit, so that as many children as possible can find a "gateway" into the topic.

Another expression of individual differences: Love languages



Knowing about love languages can help teachers recognise that children experience attention, acceptance and appreciation in different ways.

Mapping individual learning differences

To help student development, we must know students. How can we get to know them? Some tips Through

- observation in the lessons with observation sheets
- short surveys about strong and weak points, and about individual needs
- conversation and open questions
- wathing what the student likes doing, how they approach to problems, and their prefered
- ways of expression
- Following Maslow 's system
- Communicating with their parents
- Sharing experiences with other teachers

During preparation for an ECA lesson, the teahcer collects and analysis the information they have and prepares the differenciated tasks. The practical steps are:

1. prior observation and data collection
2. Self-reflection tasks
3. differentiated tasks built on Multiple Intelligence Theory
4. Individual and gropu supporting staegies
5. Communication with families and experts

While planning a lesson, we need not deviate too much from our routines: we just take the next topic. We follow our routine in fnding pput what we want to teach. Then we elaborate what social emotional questions we want to discuss in connection with oour topic.We want to find answers for the following:

- What is the group interested ina t the moment?
- What relations are there among the kids?
- What can I do for peace or for keeping the peace?
- What would I like to discuss with the students?
- What has happened lately that we can use to make the group atmosphere better?
- Are we close to a conflict?
- Who is not being well? What can we do for them?

Teachers normally deal with these questions anyaway, but here and now we create a specific space for them. Students will help us for sure.

We also have to make the first steps in connection witj Action. We need to offer very different activities and prepare the necessary tools. We can build on students' ideas, too.

Personal touch, relationships, mentoring

We have elaborated some aspects the teahcer can use during lesson preparation in order to build personal connections with students

- Take time to really listen to the child, paying attention to non-verbal signals as well
- regularly talking individually with children about personal topics and feelings;
- consciously identifying and reinforcing each child's individual strengths;
- respecting confidentiality and agreed boundaries of trust;
- developing individual goals with each child and observing changes together;
- encouraging experimentation and independence;

- accepting individual situations;
- organising shared activities;
- giving regular, constructive feedback.

Experiences of success

There are many different ways to experience success. In a class of twenty children, some will certainly enjoy writing, but even they may experience success in very different ways: one may take pride in beautiful handwriting, while another may enjoy expressing creative ideas.
The following can help support both motivation and the quality of learning:

- identifying abilities and interests;
- setting personal goals;
- taking small steps;
- providing differentiated tasks and paying attention to different areas of intelligence, especially movement;
- offering relevant tasks that can be completed at home instead of conventional homework;
- providing choices;
- giving positive reinforcement for the process as well as the result;
- providing opportunities to share experiences;
- giving supportive and honest feedback;
- supporting self-reflection and self-assessment so that learners recognise their own development;
- using games and experiential learning formats, competitions and challenges;
- creating a safe and interesting physical learning environment with alternatives to desks;
- using interactive presentations with questions, visual aids, clear structure, periodic summaries and systems thinking.

Tips for individual and differentiated learning organisation

- Recognise learners' different physical, emotional and mental needs and take them into account when creating the learning environment, while remaining aware of motivational factors.
- Observe learners' different learning preferences, strengths and current needs, and adapt tasks and tools flexibly.
- Use flexible learning pathways and learning formats by providing choices between tasks.
- Assign specific roles and rotate them regularly so that everyone can experience different roles.
- Use visual and digital tools – timers, wall calendars, shared notes and online group work – to support time management, organisation and collaboration.
- Develop shared rules formulated by the learners themselves, and use regular feedback rounds after group work.
- Organizing trainings for practicing non-violent conflict managing and cooperating in safe environment.
- The development of self-reflection with sample questions like "What went well?, "What-would you do in another way?" or through short notes or diary.
- Building a supportive and empathic teacher-student connection: tolerance of diversity, openness o student feedback, authentic communication.
- Routines and subroutines, transparent lesson structures together with variable methods, tools and safe teacher reflection.
- Cooperation with parents and educationists for supporting students with mutual share of experiences.

Integrating independent and group learning

The task expressed in this heading is one of the keys to putting a person-centred approach into practice. Here too, we have collected a number of points that can help us build our strategy.

- Clarifying learning goals: Set clear, unambiguous individual goals for learners.
- Developing conflict-management and cooperation skills: Provide parallel teaching and practice in communication and consensus-building.
- Using varied learning tools: Digital tools, presentations, posters or oral presentations can support both independent and group work.
- Providing continuous feedback: The teacher and group members regularly give one an-

other feedback on their work, strengthening self-reflection and development.

- Assigning learner roles: Allocate clear roles within the group – for example, coordinator, note-taker or timekeeper – to support organised work, individual responsibility and meaningful participation for everyone.
- Independent research and information gathering: Learners prepare material individually or in pairs, allowing them to work at their own pace and in their own way.
- Group discussion and problem-solving: Information gathered independently is shared within the group, followed by joint discussion, forming opinions and decision-making.
- Time management and scheduling: The teacher supports learners in managing their time effectively and provides sufficient time for activities.
-





- Aligning assessment criteria: Learners need to know that both individual and group performance matter, which encourages responsible participation.

ECA requires longer and more complex preparation because teaching is adapted to learners' needs. If a teacher finds enjoyment in differentiation and is strongly motivated by the individual development of learners, time spent in the classroom can become enjoyable and can lead to genuine, deep experiences and connections.

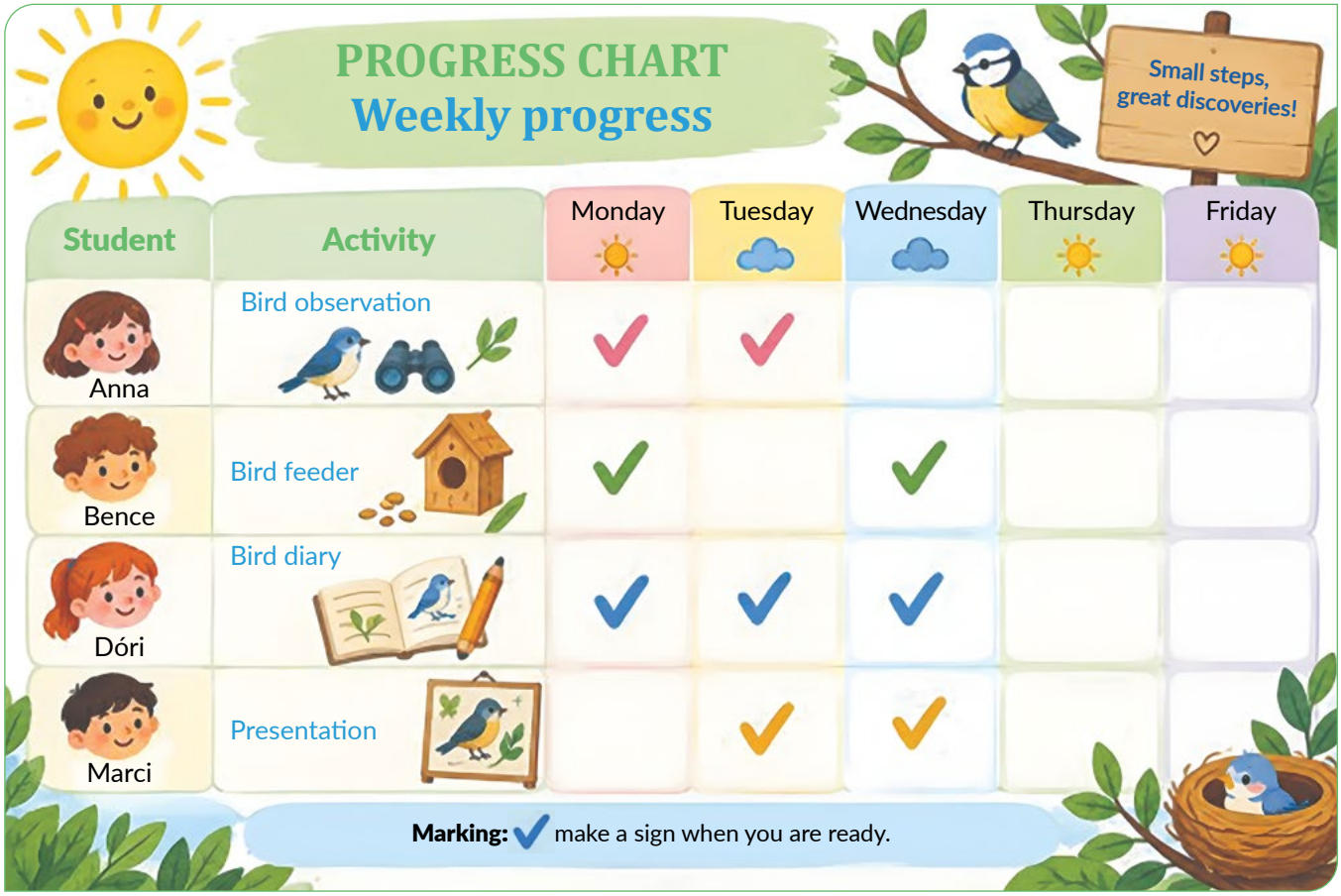
Mini self-reflection exercise

- **Think of your current class and identify three learners who are very different from one another in terms of learning. Describe in one word what makes each of them different.**
- **Decide which of them you will make one small personalised adjustment for next week: a different tool, a different amount of work, or a different type of task.**

8.
Monitoring and Assessment

Documenting individual progress

Progress chart



A Progress Chart is a visual tool. A large chart on the wall shows what each learner has undertaken and what they are currently working on. When learners complete a unit or part of a task, they can colour in the relevant cell. Because of its simplicity, the chart can easily be used to monitor weekly or monthly plans. If we notice that a learner is progressing more slowly because of their individual abilities, this difference can also be represented visually.

Individual monitoring and small-group discussions

Teachers can also monitor learning through individual or small-group discussions. Verbal feedback provides quick information about progress, while teachers' observation notes can provide a more detailed picture of learners' development over longer periods.

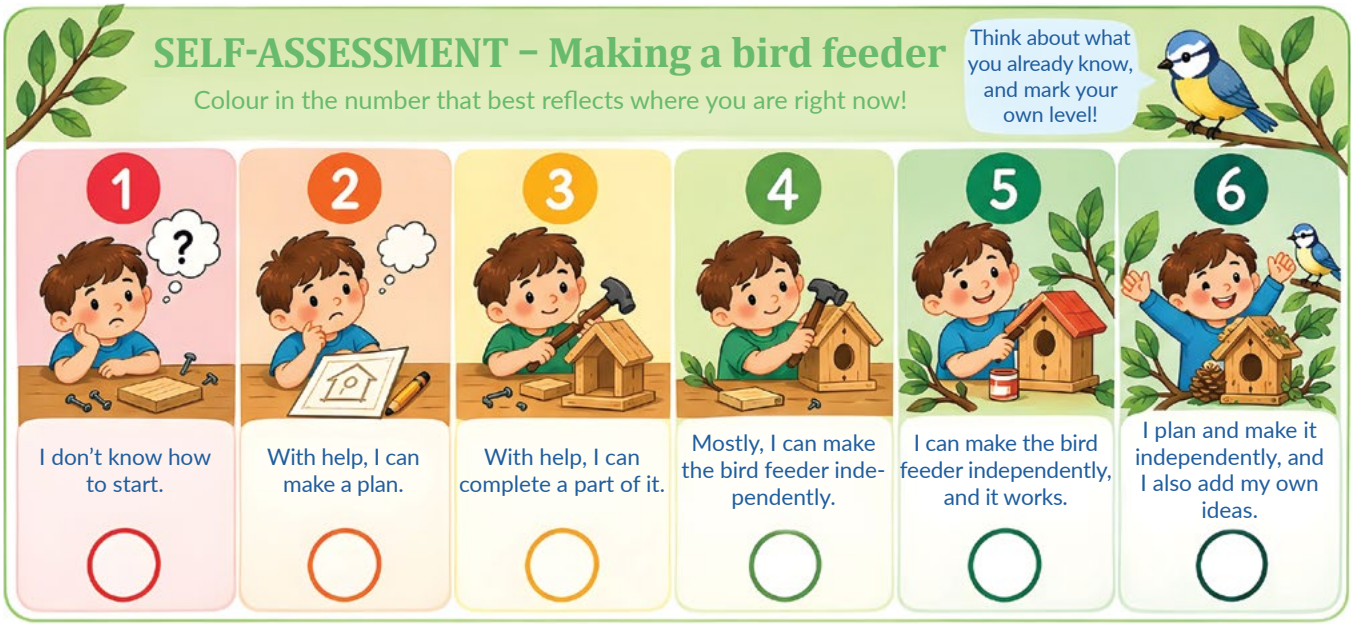
Assessment and self-assessment

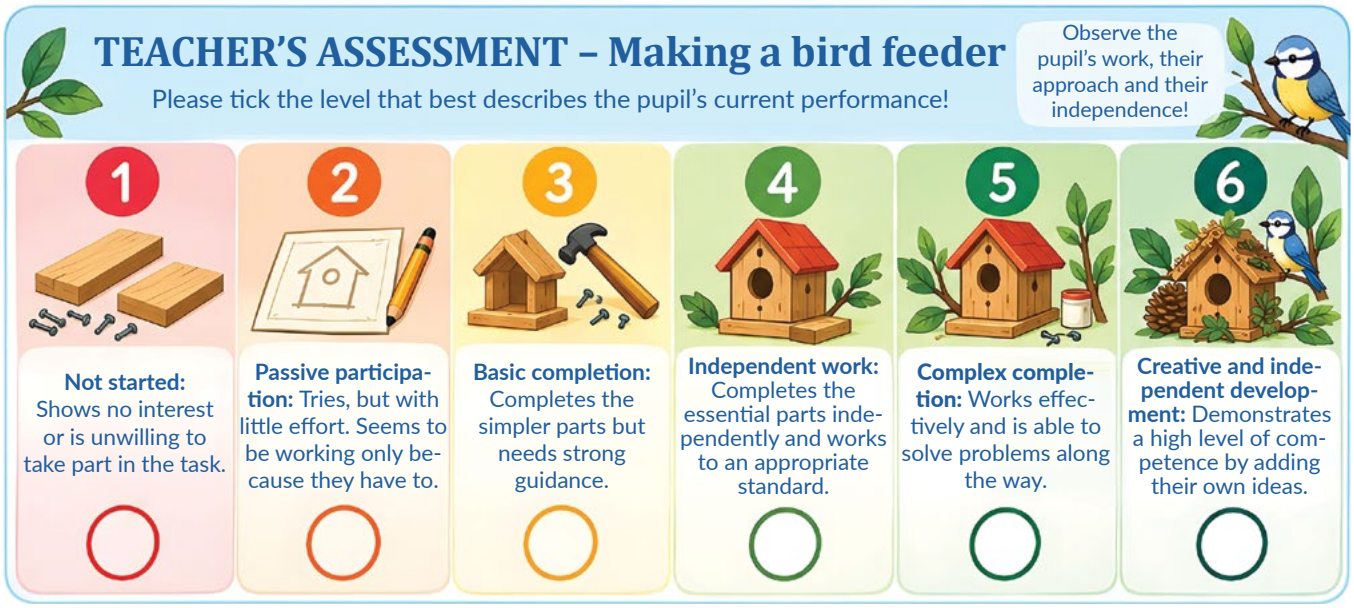
According to Carl Rogers, education is person-centred to the extent that self-assessment is at the centre of assessment. While self-assessment is not necessarily part of a traditional lesson, one of the most important aspects of assessment in an ECA lesson is how each learner sees their own progress.
Possible approaches include:

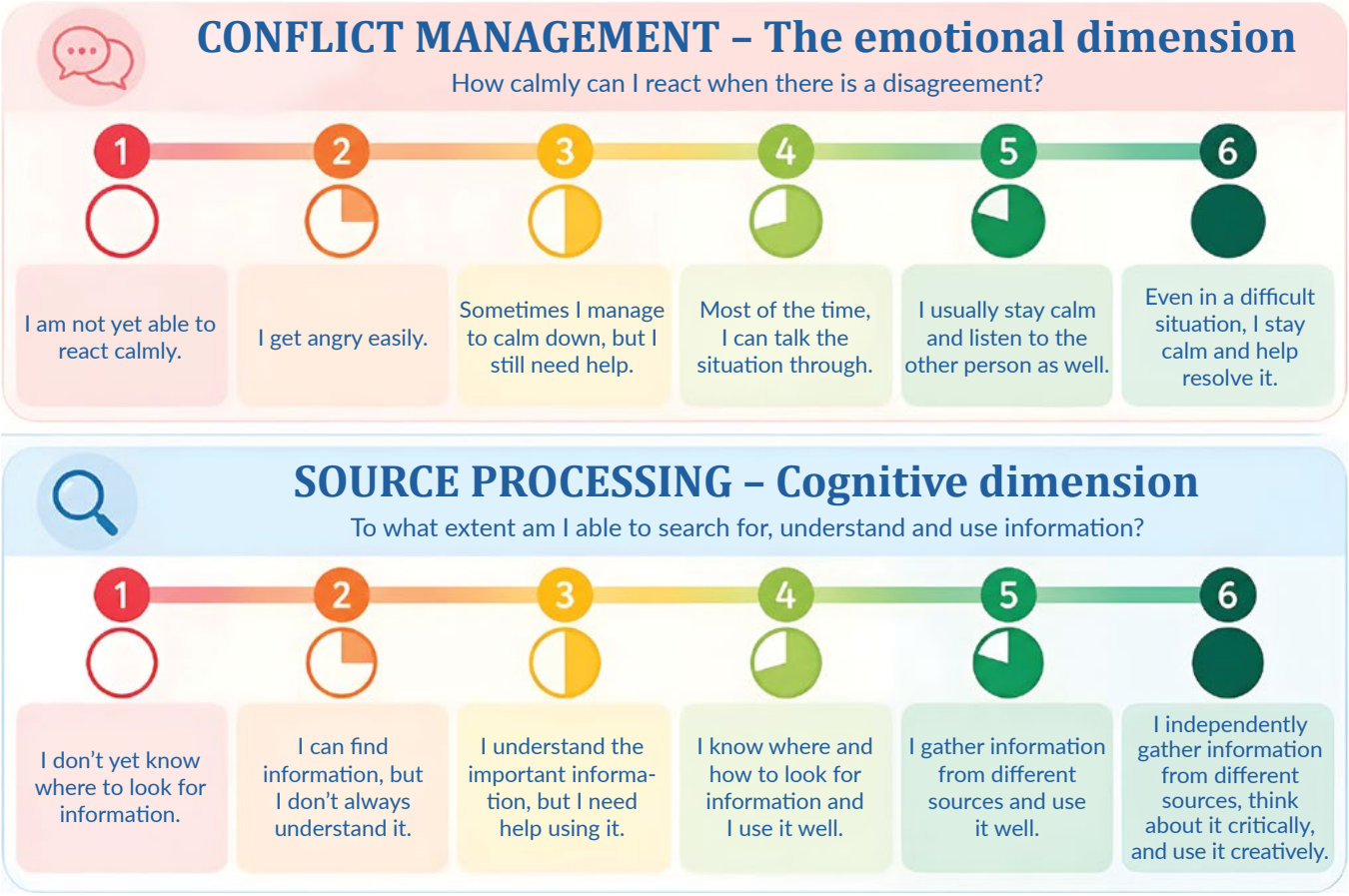
Self-assessment

- **Scaling:** Learners can assess their own performance and progress using clear scales. The teacher can complete the same scale as well. An even more person-centred approach is for children to develop the scale themselves.
- **Portfolio:** A continuously expanding collection of documents showing the learning process, achievements and development. It provides a basis for both self-assessment and teacher feedback and can also provide information for parents. It can be maintained digitally.
- **Self-assessment templates:** These can be developed for all three ECA dimensions. Care should be taken to ensure that neither the questions nor the answers become formulaic. Using different forms alongside verbal reflection – visual, movement-based, etc. – can help.

Three concrete examples show how scaling can be used, as this form is still relatively unfamiliar. Emotional, cognitive and action scales can serve as models for developing self-assessment scales in other areas. They can be used both for learner self-assessment and teacher assessment. Simpler forms can, of course, be used for any topic.







Developing a self-assessment routine

Many young children may initially be reluctant to engage in self-assessment, may not take it seriously, or may begin to give routine answers. Developing a genuine self-assessment routine takes time and patience. Teachers need to support the process consciously so that it gradually becomes a natural part of learning. Possible tools include:

1. **Using simple visual tools:** Use pictures, symbols, emojis and pictograms to help children express their emotions and assess their performance in a simple way. Develop these together with the children.
2. **Short, everyday practices and routines:** For example, at the end of a lesson or after an activity, ask children to indicate how they felt or how successful they thought the task was. One simple method is to use their arms to show satisfaction, success, etc.: arms raised means 10, arms lowered means 0.
3. **Positive reinforcement and encouragement:** After each self-assessment activity, give children positive reinforcement so that self-assessment becomes a positive experience.
4. **Shared discussions:** From time to time, sit down with the children and talk together about self-assessment.
5. **Gradually introducing self-reflection questions:** Start with simple questions such as "What was your favourite part?", and later move on to more complex ones.
6. **Tracking their own development:** Create small booklets or drawings together in which children can show their development over time, helping to strengthen the habit of self-reflection.

Feedback

Feedback can take many forms; it is difficult even to list all the ways in which we can gain an understanding of how others see us. For example:

- **Verbal feedback:** feedback from the teacher or peers, individual or group discussions.
- **Written and visual feedback:** short assessments, observation notes, scaling.
- **Learner feedback:** questionnaires about motivation, satisfaction and difficulties.
- **Formal assessment:** tasks, tests and written tests.

- **Multi-source feedback:** for example, 360-degree feedback, in which several participants in the process provide feedback. The more varied the ways in which we give feedback, the better. The same applies to assessment.

Teacher feedback and formative assessment

Formative assessment can take many forms.

A grade can also reflect several different aspects of assessment. It may take into account a range of recurring or situation-specific factors, such as meeting requirements, knowledge, the development of skills, classroom work, the way tasks are approached, diligence, and so on.

We have collected some of the factors that can influence the effectiveness of teacher assessment:

- Goal orientation: Assessment is based on clear, well-defined goals.
- Taking individual abilities or pace into account: Avoid relying exclusively on external standards.
- Quality of feedback: Feedback is specific, constructive, development-oriented and considers several aspects.
- Using a variety of assessment tools.
- Documentation and monitoring: These help strengthen awareness.
- Involving learners.
- A supportive learning environment: An atmosphere in which mistakes are accepted and learners feel safe to try.
- Regular and continuous use: Assessment is an integral part of the learning process.
- Developing the teacher's own self-awareness.

Defining assessment criteria

Ideally, assessment criteria should be developed jointly by the learner and the teacher. The learner knows what they want to assess, while the teacher knows what needs to be assessed according to the requirements of the National Core Curriculum. By developing the criteria together, the learner considers the requirements carefully, while the teacher can support the learner's work more effectively. Naturally, all of this needs to happen before the work begins.

What should we consider when defining assessment criteria?

- **Assessment criteria should be developed jointly by the learner(s) and the teacher.**
- **Goals should be clear and measurable:** everyone involved should know what the expected outcome is.
- **Different levels should be defined:** these should always be phrased in an encouraging way.
- **Assessment should include several criteria:** saying that the use of sources was not acceptable tells us much less than breaking the assessment down into smaller components, such as the quality and number of sources used, formal elements, content, language quality and argumentation.
- **Several levels should be defined for each criterion** so that results can be compared.
- **Relevance and alignment:** make sure that the criteria are aligned with the learning objectives and are relevant to the task.
- **Simplicity and clarity:** formulate the criteria clearly; they should not be complicated or ambiguous.
- **Review based on experience:** together with the learners, consider whether anything needs to be changed in light of experience.

Self-assessment and grading

It is up to the teacher to decide to what extent children's self-assessment becomes part of their mid-year or end-of-year grades. Embedding self-reflection and self-assessment in the learning process is already a significant step. But it is even more valuable when a young child experiences that their own view of their performance actually influences the grade recorded in their report card. The agreement between teacher and learner on how self-assessment is incorporated into the final grade can take place at several levels:

1. The teacher listens to the learner's self-assessment and responds to it, but does not take it into account when determining the grade.
2. The teacher listens to the learner and takes their self-assessment into account when determining the grade.
3. The grade is agreed jointly through an assessment conversation.
4. The teacher accepts the grade the learner gives themselves.

On what basis do we accept a student's self-assessment?

Fortunately, there is also movement at an institutional level away from assessment expressed exclusively through grades. In many places, developmental written assessments are already used in the lower classes.

It is important to recognise that the different elements of person-centred education are interconnected. They form a system:

1. identifying individual needs – preferences and needs;
2. alternating between independent and group work;
3. measurement and monitoring, including a Progress Chart with E/C/A columns;
4. individual monitoring and small-group reflection;
5. fine-tuning using checklists.

Mini self-reflection exercise

- **Consider how often you ask children:** "How would you assess your work today?" – never, sometimes or often.
- **Write down two simple self-assessment questions that you will try next week,** for example: "What went well today?" and "What would you do a little differently next time?"

9.
Adapting and Creating Your Own ECA Materials

There is no single correct way to apply ECA. The same ECA idea may work differently in a first-grade and a third-grade class, with a whole class or a smaller group, in a short lesson or a longer learning unit. Practical ECA materials therefore need to be adapted to the particular children, learning objective and situation. The examples and tools in this Handbook are starting points. What matters is that the person-centred approach of ECA remains intact after adaptation.

When using an existing ECA activity, it is worth first identifying its essential element. What do we definitely want to preserve? Is it the emotional connection? An interesting question? The opportunity to choose? Direct experience? Creating something? Collaborative problem-solving? Once we have identified this core element, the external form of the activity can be changed more freely.

An ECA material that we have created or adapted truly becomes our own when we try it out. Based on our experience, we can modify it and then try it again. Over time, we can develop our own ECA toolkit: a collection of activities, questions, tasks and forms of assessment that we know from experience work with our own learners.

ECA can also be understood as a pedagogical structure built on a foundation of shared values. On this foundation stand the supporting elements of Emotion, Cognition and Action: a wide range of attitudes, skills and methodological decisions that cannot – and need not – be listed exhaustively.

Assessment and self-reflection are the binding material that holds these elements together. They are present at every level and continuously shape the whole structure. Within this metaphor, the teacher is the architect: the person who responsibly shapes the relationship between the different elements in order to create a safe, personal and creative space for children. In such a space, motivation is not an external tool but a natural consequence of learning – autonomy, competence and safety – for both children and teachers.



10.
Short lesson plans from practice

This chapter is included in the Handbook to make the ECA approach visible in practice. The lesson plans below are not "ready-made recipes", but visual models. They are not intended to be adopted without modification; rather, they show how a learning situation can be organised according to the ECA logic. Each lesson plan follows the same structure: emotional attunement, shared thinking and active engagement, interwoven throughout the learning process. The examples come from

Example 1

The Earth (LP1)

ECA Interpretation

- **E – Emotion:** The topic of the Earth connects with personal experiences – excursions, home environment and nature – and therefore naturally creates involvement and emotional safety.
- **C – Cognition:** Through observation and comparison, children create their own categories while developing basic concepts about the planet and its environment.
- **A – Action:** Creating something – a model, drawing or shared visual representation – provides opportunities for active participation and allows for different paces and different solutions.

Teacher focus: The emphasis is not on producing the "correct" representation, but on making the thinking process visible.









Example 2

Volcanoes (LP 3)

ECA Interpretation

- **E – Emotion:** Volcanoes can evoke strong emotional reactions – excitement, fear and curiosity – which can consciously be incorporated into the learning process.
- **C – Cognition:** By observing and discussing phenomena, children explore cause-and-effect relationships and formulate their own questions.
- **A – Action:** Modelling and experimentation deepen understanding through action and provide space for individual exploration.

Teacher's focus: Acknowledging and interpreting emotional reactions is at least as important as the science content. The volcano can also be used as an analogy for understanding anger and strong emotions.







Example 3

The Human Body (LP 6)

ECA Interpretation

- **E – Emotion:** The topic of the body offers a direct connection to self-awareness.
- **C – Cognition:** Children use their own bodily experiences to understand basic bodily functions and relationships.
- **A – Action:** Large-scale collaborative creation and movement-based elements provide opportunities for active participation and cooperation.

Teacher focus: The activity supports self-reflection rather than the memorisation of details.







The materials presented here are not intended merely to transmit knowledge; they also reflect an approach to applying ECA in pedagogical practice. The images and templates allow teachers to try out the methods in their own teaching environments and adapt them to the individual needs of their learners.

Throughout the chapter, the processing of learners' experiences, reflective practice and creative problem-solving have been emphasised. On the one hand, these help consolidate knowledge; on the other, they strengthen children's autonomy, motivation and emotional safety.

Overall, this chapter forms a bridge between pedagogical theory and everyday practice. It provides inspiration for applying a person-centred approach and demonstrates how templates, visual aids and project-planning tools can be integrated in ways that allow each learner's abilities and individual pathway to emerge.

The teacher is not simply a guide, but a partner in the learning process, creating opportunities for children's creativity, self-confidence and independent thinking to develop.

